Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/20/2006
PI Name:	Charles, Harry K. Ph.D.		
Project Title:	Advanced, Multiple Projection, Dual Energy X-ray Absorptiometry (AMPDXA) Scanning System		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20723-6099	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	03/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Beck, Thomas (Johns Hopkins Medical Institution)		
Grant/Contract No.:	TD.002.03		
Performance Goal No.:			
Performance Goal Text:			
	The purpose of the advanced, multiple projection, dual energy x-ray absorptiometry (AMPDXA) scanner system project is to design, built, and test a precision scanner for monitoring the deleterious effects of weightlessness on the human musculoskeletal system during prolonged spaceflight. The instrument uses dual energy x-ray absorptiometry principles and is designed to measure bone mineral density (BMD), decompose soft tissue into fat and muscle, and derive bone structural properties (cross-sections, moments of inertia). Such data permits assessment of microgravity effects on bone and muscle and the associated fracture risk upon returning to planetary gravity levels. Ultimately designed for use on the space station or other long duration space missions, the AMPDXA will weigh less than 45 kg in its flight version and be capable of near real-time monitoring of bone loss as well as the effects of countermeasures. The AMPDXA will be able to detect changes of less than 1% in bone mass and geometry and less than 5% in muscle mass. Specific tasks underway or planned for the near future include:		

Task Description:	1. Human testing with the Clinical Test System to develop the final measurement parameters (e.g. dose, duration of X-ray pulses, final filters,operational screnarios, etc.) 2. Refinement of software algorithms for the extraction of soft tissue. 3. Developing a solution to the two-dimensional versus three-dimensional reconstruction problem. 4. Refinement of algorithms for relating bone strength and structure to the risk of fracture. 5. Software (and instrumentation) refinements to allow the collection of radiographs for diagnosis of injury and disease. 6. Design of a lightweight power supply.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In microgravity, astronauts lose between 1.0 and 1.5% of their bone and muscle mass per month, primarily in the vertebrae, pelvis, and proximal femurs. This loss should be compared to the loss from these sites of approximately 0.8 to 1.3% per year in postmenopausal women. In addition, recent studies indicate that astronauts may be at a greater risk of fracture for the same loss of bone mass. Thus, it is important to not only determine bone mass loss and muscle atrophy, but also to determine the geometrical configuration of the bone structure. From the structural analysis of each astronaut?s bones, an estimate of the specific load-bearing capability of their musculoskeletal system can be made and forewarn of any danger. The AMPDXA is particularly suited to this task since it provides a highly accurate quantitative means to monitor tissue loss and structure, which then can be used to gauge risk of serious consequences (fracture) and assess the effectiveness of countermeasures.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 12/16/2008)